

Question.

Implement a stack using a circular linked list.

Question

Given a list L of integers $a_1, a_2, \dots, a_n$ and an integer M , describe an algorithm that finds the largest subset of L whose sum is at most M . Your algorithm should run in linear time.

Sample Input:

$L = \{2, 5, 1, 3, 4\}$

$M = 8$

Sample Output:

Largest subset = $\{2, 1, 5\}$

Subset Size = 3

Sum = 8

Question

You are given an array:

5, 4, 2, 7, 9, 6

Convert it into a sorted array with the minimum number of operations. The only operation allowed is decrement (decrease an element by 1).

Sample Input:

5 4 2 7 9 6

Sample Output:

2 2 2 6 6 6

Explanation:

Total decrement operations = 9

Question

Find the minimum positive integer from a given array of n numbers.

Sample Input:

4 -2 0 1 3 -1 2 6 8 -5

Sample Output:

1

Question

Find the minimum missing positive integer from the given array in $O(n)$ time complexity.

Sample Input 1:

2 3 4 -1 7 6 0

Sample Output 1:

1

Sample Input 2:

1 2 3 4 -1 7 9 0

Sample Output 2:

5

Sample Input 3:

-2 -1 0 1 2 4 5 6 11

Sample Output 3:

3

Question

You are given an unsorted array of integers, and the length of the array is very large. Write an efficient algorithm to retrieve the top N largest integers. The result need not be sorted.

Sample Input:

Array = {10, 3, 5, 22, 18, 9, 15, 30}

N = 3

Sample Output:

30 22 18

(Note: The output may be in any order since sorting is not required.)